

Insect Ecology

Zool. 503
B. Ecology

Definition of Ecology

According to the Ecologists.....

Haeckel	1860s	The total relations of an organism to its organic and inorganic environment
Richards	1890s	Living in harmony with the environment, first including the human species
Elton	1920s	Scientific natural history
Odum	1960s	The study of structure and function of nature, including the human species
Andrewartha	1960s	The scientific study of the distribution and abundance of organisms
Krebs		The scientific study of the interactions that determine the distributions and abundance of organisms
Molles	1990s	The study of relationships between organisms and the environment
Eltis	2010s	Life in context
Pope Francis	2015	The relationship between living organisms and the environment in which they develop

Haeckel	1860s	The total relations of an organism to its organic and inorganic environment
Richards	1890s	Living in harmony with the environment, first including the human species
Odum	1960s	The study of structure and function of nature, including the human species
Krebs		The scientific study of the interactions that determine the distributions and abundance of organisms
Pope Francis	2015	The relationship between living organisms and the environment in which they develop

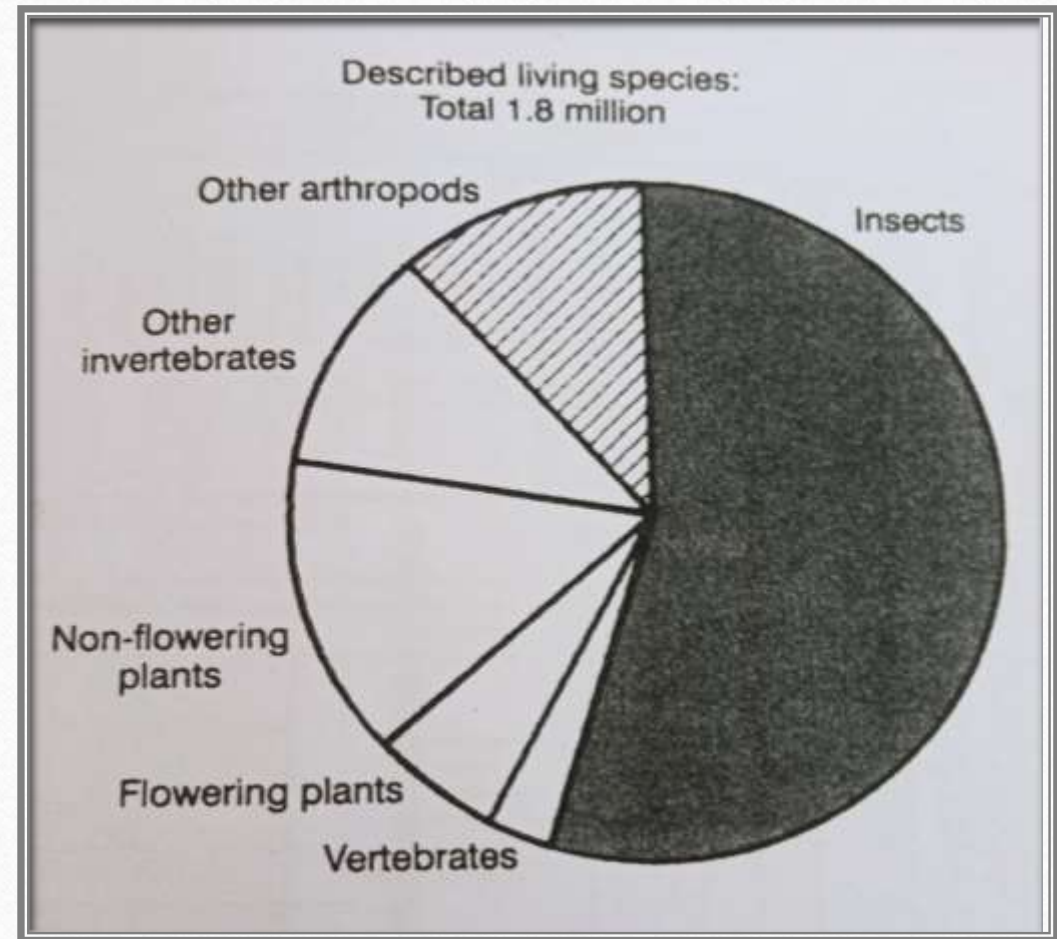
Definition of Insect Ecology

- ➡ The insect ecology is the scientific study of how insects, individually or as a community, interact with the surrounding environment or ecosystem.
- ➡ Insect ecology is the branch of entomology that focuses on the interrelationships between insects and their environment.
- ➡ Insect Ecology is science of insect in relation to their environment.

Why Insect Ecology is Important?

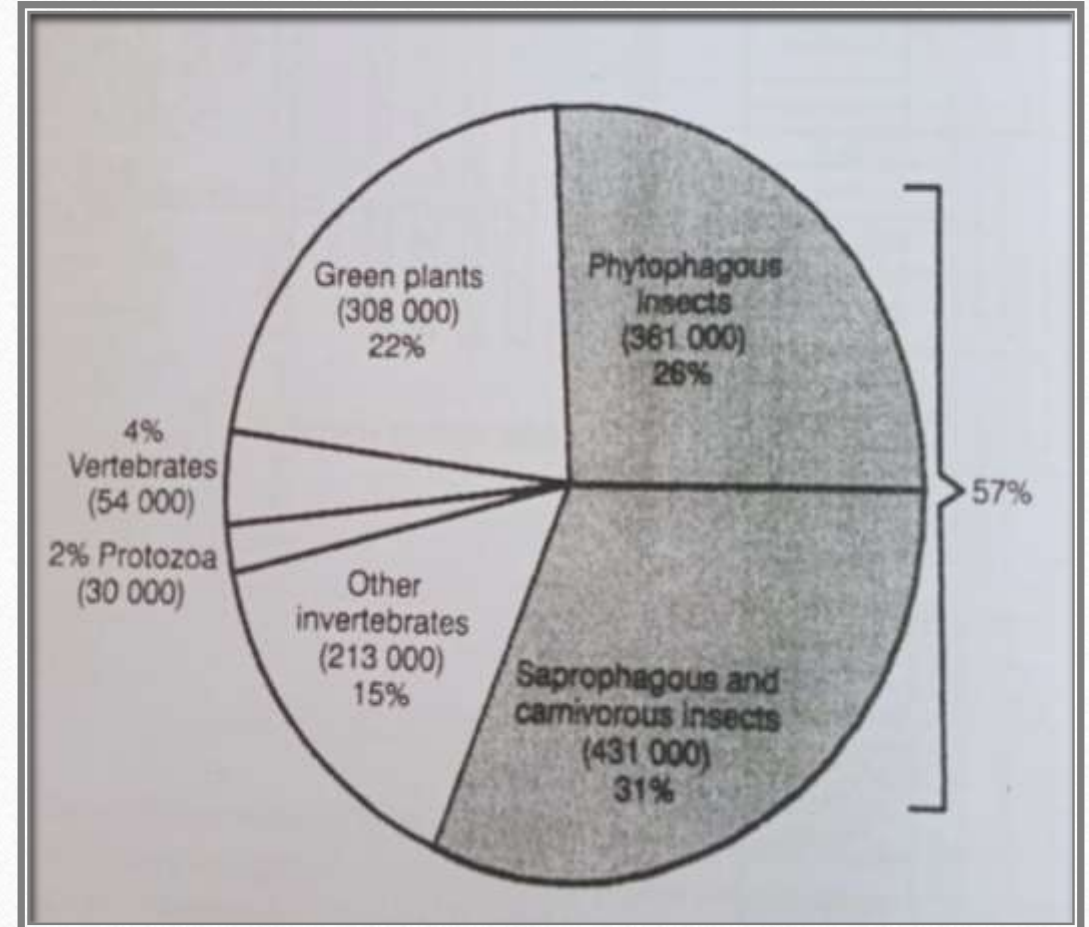
Percentage of all described species in each major taxon out of the estimated 1.8 million species described. From Stork (1988); with permission from the Linnean Society of London

Insects are occupied with almost 60% of species (Stork, 1988)



Role of Insects in Food Chains and Food Webs

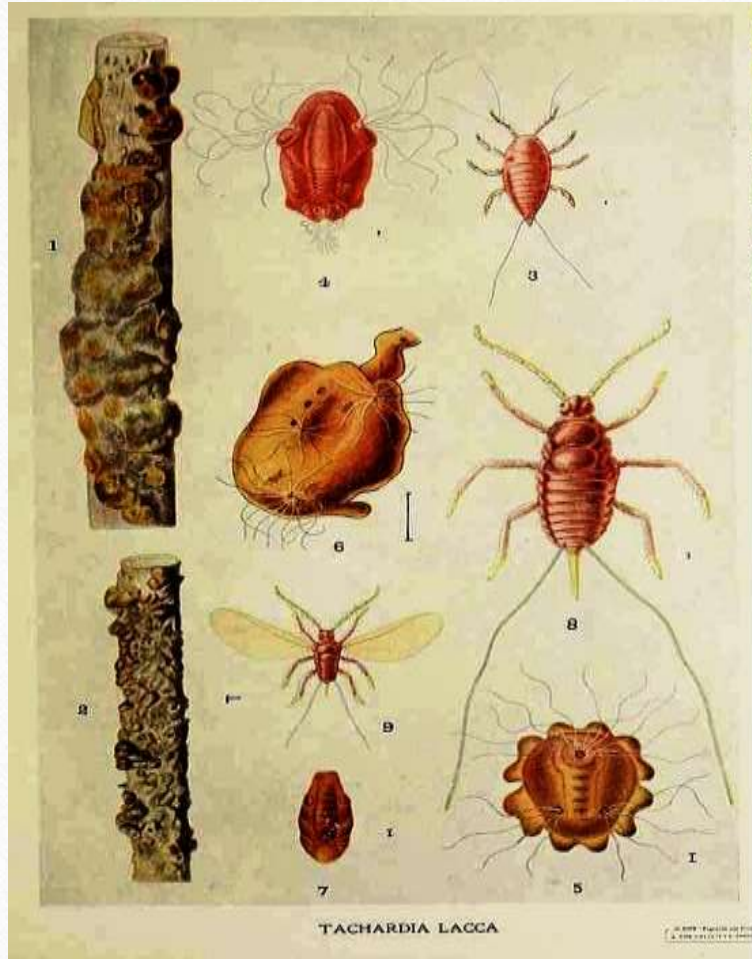
- ▶ **Phytophagous Insects** – 26% - They are converting plant biomass to animal energy
- ▶ **Saprophagous and Predaceous insects** – 31% - They are forming important components of terrestrial and freshwater food webs
- ▶ ▶ It means 57 % of terrestrial species are involved as intermediaries in food webs (Second and Third trophic levels)



Beneficial Insects

- ▶ Pollinators
- ▶ Biological Pest controls (Predators and Parasitoids)
- ▶ Tilling the Soil to recycling the corpses of plants and animals
- ▶ Forensic Entomology
- ▶ Important role in Food Chain
- ▶ Commercial Product like Honey, Wax, Silk, Shellac (*Laccifer lacca*) and Cochineal(*Dactylopius coccus*- a scale insects) and their galls (which provide tanning that are the basis for inks)
- ▶ Human Food

Shellac – Female Lac Bug



Cochineal



As Pest

- ▶ Agricultural pest
 - ▶ Crop pest
 - ▶ Stored grain pest
- ▶ Veterinary Pest
- ▶ Human Disease
- ▶ Timber pest

Other Importance

- ▶ **Good environmental indicator** (Ex: Lime Butterfly)
- ▶ **Bio-indicator for aquatic ecosystem** (Ex: Dragonfly & Damselfly)
- ▶ **Economic Damage** (Ex: Grasshoppers)
- ▶ **Cultural Services** (Ex: Monarch Butterfly)

ASSIGNMENT

Make a brief account on the importance of insects in the Ecological, Biological and Economical aspects.

Thank You